

Experiment 02: Hardware Configuration & Interfacing Schematics

1. Arduino UNO R3 Hardware Architecture & Pinout

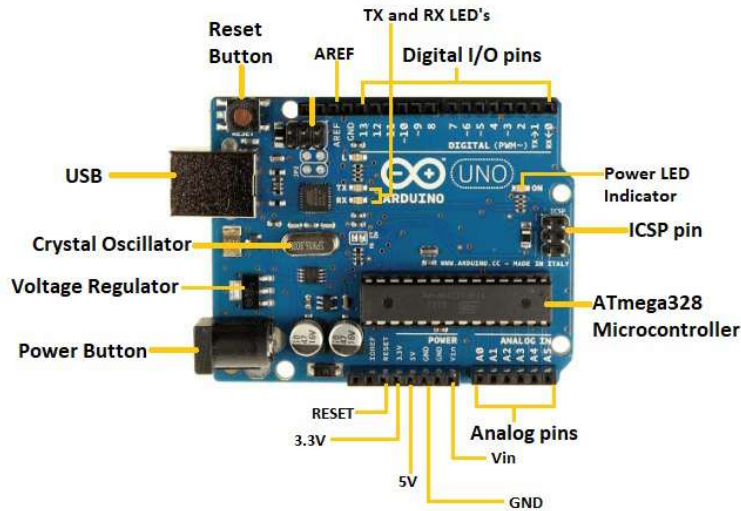


Figure 2.1: Arduino UNO R3 Hardware Architecture and Pin Configuration Layout.

2. Dual Alternating LED Interfacing Layouts

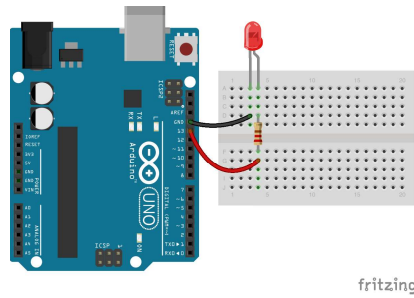


Figure 2.2: Breadboard Fritzing Wiring Layout for Dual LED Interfacing.

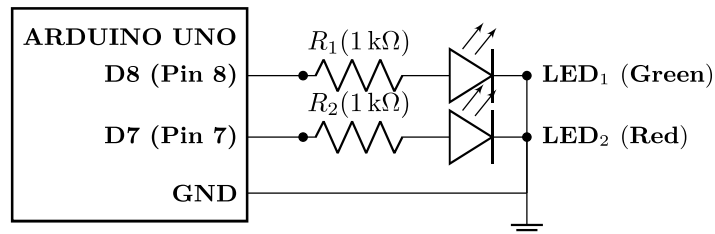


Figure 2.3: CircuitiKikZ Dual-LED Interfacing Circuit Schematic.

3. Hardware Pin Wiring Interconnection Matrix

Arduino Terminal	Breadboard Component	Component Terminal	Signal Function
Digital Pin 8 (D8)	Resistor R_1 (1 k Ω)	Lead 1	GPIO Push-Pull Output Channel 1
Resistor R_1 (1 k Ω)	LED ₁ (Green)	Anode (Long Leg, +)	Current-Limited 5V Drive ($I_{F1} \approx 3.0$ mA)
LED ₁ (Green)	Common Ground Rail	Cathode (Flat Notch, -)	System Ground Return Path
Digital Pin 7 (D7)	Resistor R_2 (1 k Ω)	Lead 1	GPIO Push-Pull Output Channel 2
Resistor R_2 (1 k Ω)	LED ₂ (Red)	Anode (Long Leg, +)	Current-Limited 5V Drive ($I_{F2} \approx 3.0$ mA)
LED ₂ (Red)	Common Ground Rail	Cathode (Flat Notch, -)	System Ground Return Path

Table 2.1: Pin Wiring Interconnection Matrix for Experiment 2.

Experiment 03: DHT11 Sensor Protocols & Circuit Schematics

1. DHT11 Single-Bus (1-Wire) Communication Handshake

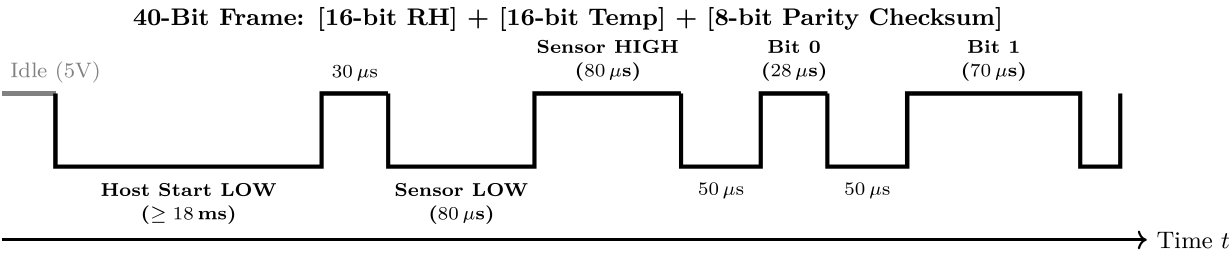


Figure 3.1: DHT11 Single-Bus 1-Wire Handshake and Bit Timing Sequence.

2. DHT11 Sensor Breadboard Wiring & Interfacing Circuit Schematic

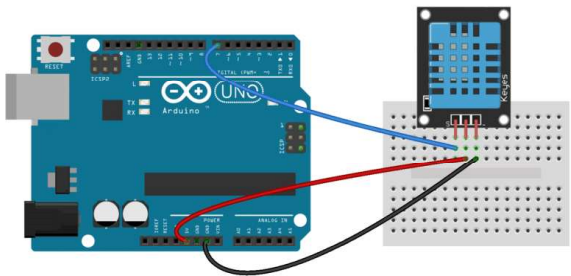


Figure 3.2: DHT11 Breadboard Fritzing Wiring Layout.

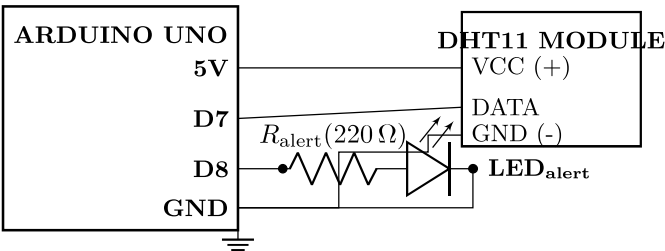


Figure 3.3: DHT11 Sensor & High-Temperature Alert LED Circuit Schematic.

3. Hardware Pin Wiring Interconnection Matrix

Arduino UNO Pin	External Component	Component Terminal	Functional Role
+5V	DHT11 Module	Pin 1 ($V_{CC}/+$)	Sensor DC Power Supply (+5V)
Digital Pin 7 (D7)	DHT11 Module	Pin 2 (DATA / OUT)	Bidirectional 1-Wire Telemetry Line
GND	DHT11 Module	Pin 3 (GND / -)	Common System Ground (0V)
Digital Pin 8 (D8)	Resistor R_{alert} (220 Ω)	Lead 1	High-Temperature Alert Output Trigger
Resistor R_{alert}	Alert LED (Red)	Anode (Long Leg, +)	Current-Limited LED Driver ($I_F \approx 13.6$ mA)
Alert LED	Arduino GND Rail	Cathode (Flat Notch, -)	Ground Return Path

Table 3.1: Hardware Wiring Interconnection Matrix for Experiment 3.